H

RAFFLES INSTITUTION

2013 Year 6 Preliminary Examination
Higher 1

CANDIDATE NAME	

CIVICS GROUP	1	3	S	0	3		

INDEX NUMBER						

BIOLOGY **8875/02**
Paper 2 Core paper **16th SEPTEMBER**
2013

2
hours

Additional materials: Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **either ONE** question.

At the end of the examination, **hand in your essay SEPARATELY.**
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**/12**
2	**/12**
3	**/8**
4	**/8**
Section B	
5 OR 6	**/20**
Total	**/60**

This document consists of 10 printed pages.

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Section A

Answer **all** the questions in this section.

Raffles Institution
Internal Examination

1 **Fig. 1.1** shows the sequence of bases in a section of a gene coding for the first seven amino acids of a polypeptide chain.

5' A A A C T G T T T C T G C T G G A A C A T 3'

Fig. 1.1

(a) **(i)** Methionine is coded by the initiation codon. On **Fig. 1.1**, label, the 3' and 5' end of the DNA template strand. [1]

(ii) List the corresponding codons in the mRNA transcribed from the template in **Fig. 1.1**.

AUG UUC CAG CAG AAA CAG UUU

A: 3' UUU GAC AAA GAC GAC CUU GUA 5' [1]

The polypeptide was hydrolysed to release the first seven amino acids. It contained four different amino acids. The number of each type obtained is shown in the **Table 1.1**.

Amino acid	Number present
Phe	2
Met	1
Lys	1
Gln	3

Table 1.1

(iii) Using information in **Table 1.1**, work out the order of amino acids in the polypeptide.

met ***phe*** ***gln*** ***gln*** ***lys*** ***gln*** ***phe***

[1]

(b) **(i)** Explain why single base deletions in the DNA sequence is more likely to produce non-functional proteins than single base substitutions.

1. ***deletion results in frameshift***
2. ***Change in sequence of codons thus /causing all amino acids downstream of mutation to change***
3. ***substitution usually cause change in 1 aa/1 codon;***
4. ***or may result in silent mutation that results in no change in aa (if it occurs at 3rd base as a result of degeneracy of genetic code) Or aa has similar property***
5. ***Substitution may not lead to conformational change of protein but deletion***

mutation will. WTTE [3]

(ii) Describe the effect of replacing guanine (underlined in **Fig.1.1**) with adenine.

1. AAG → AAA / UUC → UUU
2. Due to degenerate code (as more than one codon may code for the same aa);
3. ***still code for phe / no change in aa (silent mutation);*** [2]

(c) Explain how the structure of the nuclear envelope facilitates translation.

1. The presence of *nuclear pores ;***
2. Which provide the passage for mature mRNA to move out of nucleus to ribosome in cytoplasm for translation;
3. Ribosomal subunits to move out of nucleus to cytoplasm to form ribosome /tRNA move out of nucleus to cytoplasm to undergo amino acid activation;
4. Ribosomal proteins to move into nucleus from cytoplasm to form ribosomal subunit;

Pt 1 must be in
Mark for 2 to 4 is awarded for direction of movement and molecule [2]

(d) A variety of related proteins can be produced from a gene. State the process and explain how the same gene can produce different protein products.

1. *Alternative splicing**;
(Spliceosome binds to intron-exon boundaries in the pre-mRNA)
Resulting in different exons being spliced together in different ways leading to formation of many different mature mRNA resulting in different proteins translated

[2]

[Total : 12]

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2 An **absorption** spectrum is a graph of the absorption of different wavelengths of light by a photosynthetic pigment.
An **action** spectrum is a graph of the rate of photosynthesis at different wavelengths of light.
Fig. 2.1 shows the absorption spectra of chlorophyll a and chlorophyll b as well as an action spectrum.

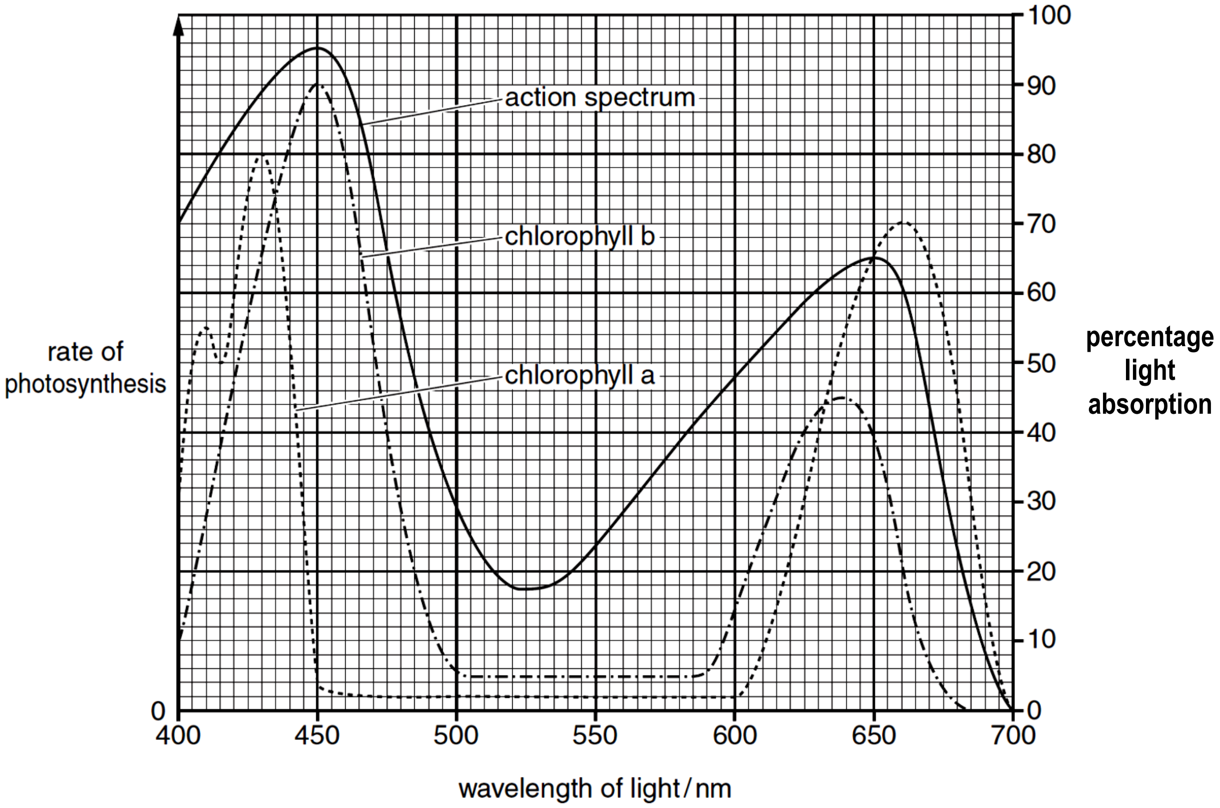

Fig. 2.1

(a) With reference to **Fig. 2.1**,

(i) compare the **absorption** spectra of chlorophyll a and chlorophyll b. [2]

Differences

1. **For percentage of light absorption, chlorophyll a <u>peaks at 430nm</u> with <u>80%</u> of light absorption while chlorophyll b <u>peaks at 450nm</u> with a <u>higher</u> <u>90%</u> light absorption;**
2. **Chlorophyll a <u>peaks at 660nm</u> with <u>much higher</u> light absorption of <u>70%</u> while chlorophyll b <u>peaks at 635–640nm</u> with <u>45%</u> light absorption/ peak of light absorption for chlorophyll a at 660nm.**

Peaks with data with max wavelength;

The trough differences with data;

Similarities

1. **Both shows higher light absorption of <u>blue light (400–500nm)</u> and <u>red light (600–700nm)</u> than other wavelengths;**
2. **Both <u>peaks in blue</u> light are <u>higher than red</u> light;**
3. **Both have <u>little absorption</u>, between <u>500–600nm / in green light</u>;**
   Or

Little light absorption for chlorophyll a between 450–600nm and for chlorophyll b between 500–600nm ;

1 difference and 1 similiarity

(ii) explain the shape of the **action** spectrum. [2]

1. **Peak(s) of action spectrum/ highest rate of photosynthesis, correspond to absorption peak(s);**
2. **Light absorbed is used for photosynthesis thus the higher percentage of light absorption leads to higher rate of photosynthesis in red and blue light ;**
3. **The region between 500 to 600nm/in the middle is not an exact match between absorption and action spectra**
4. **due to the role of carotenoids and other accessory pigments in light absorption that pass the energy to the reaction centre(link to pt 3)**

(b) The light-dependent stage of photosynthesis takes place on the thylakoids of the chloroplast. **Fig. 2.2** represents a section through a thylakoid to show some of the components involved in the light-dependent stage.

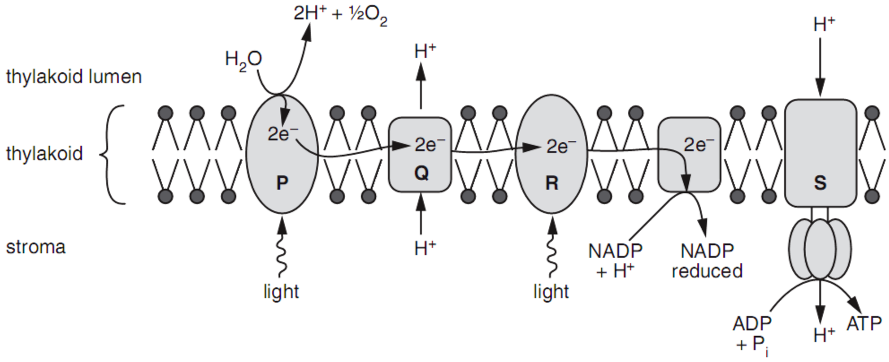

Fig. 2.2

(i) Identify the structure labelled **S**.

S : **ATP synthase/ stalked particle** [1]

(ii) Atrazine is a widely used herbicide (weed killer). It binds to a chloroplast protein involved in electron transfer between photosystems.
Explain how atrazine works as an effective herbicide. [3]

1. **Electron cannot be passed down the ETC;**
2. **No reduced NADP/ NADPH + H^+ produced;**
3. **Resulting in no / reduced, pumping of protons across the thylakoid membrane and thus no proton gradient / proton motive force**
4. **Chemiosmosis does not occur thus no ATP is produced**
5. **no, carbon fixation / Calvin cycle / light independent stage thus no (hexose) sugars / organic molecules, formed for respiration and plant dies ;**

(c) In an investigation, mammalian liver cells were homogenised (broken up) and the resulting homogenate centrifuged. Samples of the complete homogenate and samples with only mitochondria were incubated with:

1 glucose
2 pyruvate
3 glucose and cyanide
4 pyruvate and cyanide

Cyanide inhibits oxidative phosphorylation.

After incubation the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in **Table 2.1**.

	Sample of homogenate			
	Complete		Only mitochondria	
	Carbon dioxide	Lactate	Carbon dioxide	Lactate
1 glucose	√	√	X	*X*
2 pyruvate	√	√	√	*X*
3 glucose and cyanide	*X*	√	X	*X*
4 pyruvate and cyanide	*X*	√	X	*X*

***X* = absent √ = present**

Table 2.1

(i) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but **not** when they are incubated with glucose. [2]

1. **Pyruvate can enter the mitochondrion but glucose cannot**
2. **Glucose needs to be converted to pyruvate via glycolysis that takes place only in the cytoplasm/ cytosol;**
3. **Carbon dioxide is produced by *decarboxylation** of pyruvate via link reaction / of acetyl CoA in Krebs cycle in mitochondrion;**

(ii) With reference to **Table 2.1,** explain the difference in the results when the complete cell homogenate is incubated with **2** and **4**. [2]

1. **Carbon dioxide is produced only when incubated in 2;**
2. **oxidative phosphorylation cannot take place and reduced NAD is not oxidised/ NAD is not regenerated AW ;**
3. **Krebs cycle /link reaction stops so no decarboxylation takes place ;**

[Total : 12]

3 In cats, the gene controlling cat colour is sex-linked. This gene has two alleles.

In a particular colony of cats the coats of the males were either black or ginger and those of the females black, ginger or tortoiseshell (a patchwork black, ginger and cream).

(a) Using appropriate symbols, show the genotypes and their corresponding phenotypes of the different coats of the male cats mentioned above.[1]
X^BY = black male
X^GY = ginger male

(b) Using a genetic diagram, determine the phenotypes expected in the progeny of a cross between a black female and a ginger male.[4]

Parental phenotypes : black female x ginger male) [1]
Parental genotypes : X^BX^B X^GY)
Gametes [1] X^B X^G Y

Punnett Sq [1]

	X^B	X^B
X^G	**X^B X^G** Ginger female	**X^B X^G** Ginger female
Y	**X^B Y** **Black male**	**X^B Y** **Black male**

Offspring genotypes : 1 X^BX^G : 1 X^BY) [1]
Offspring phenotypes : 1 tortoishell female : 1 black male)

(c) In a litter of kittens, half the no. of females were tortoiseshell and the other ½ were black; half the no. of males were black and the other half were ginger.

What were the colours of the parental male and female? Explain your answer.[3]

1. Colour of parental male : **black** (X^BY) Colour of parental female : **tortoiseshell** (X^BX^G) [1]

Explanation :

2. Female parent contributed one of the **X chromosomes** to the male offspring, their mother must be **tortoiseshell** to have black and ginger male offspring
3. **black female offspring** indicates that male parent contributed **X^B allele thus male parent is black** (X^BY) [1]

[Total : 8]

4 To create transgenic maize which can produce the *Bacillus thuringiensis* toxin, the recombinant plasmid shown in **Fig. 3.1** was constructed. The recombinant plasmid also contains the gene for phosphomannose isomerase (PMI), an enzyme which allows plants cells to utilize mannose as a carbon source.

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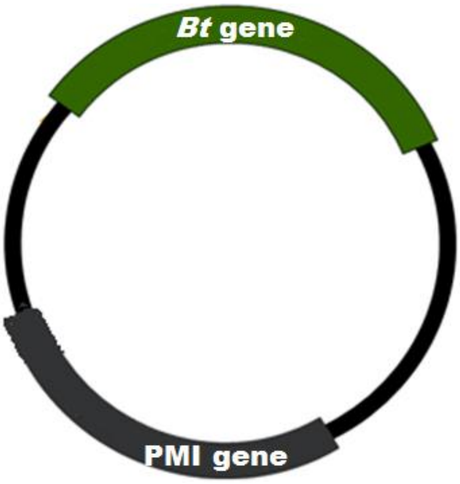

Fig. 4.1

(a) Antibiotic or herbicide resistance genes are commonly included in the plasmid used for creating transgenic plants.

Explain why such genes were not needed in this case. [2]

1. The PMI gene serves as the ***selectable marker**** instead;
2. In medium with only mannose, transformed plant cells would have the PMI gene and survive (ORA);

To produce transgenic maize using the plasmid shown in **Fig. 4.1**, the procedure shown in **Fig. 4.2** was carried out.

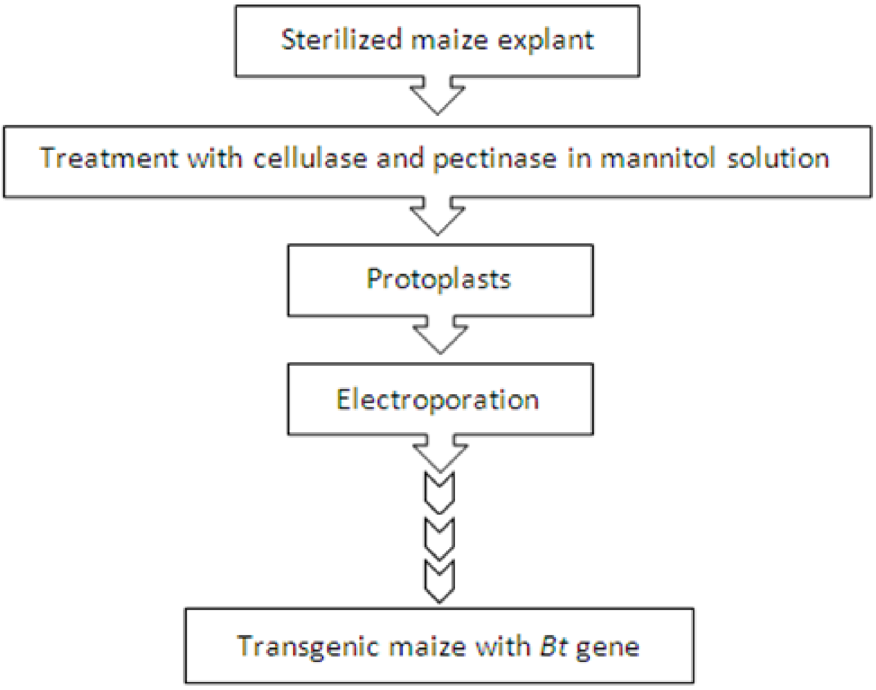

Fig. 4.2

With reference to **Fig. 4.2**,

(b) **(i)** explain why the sterilized explant was treated with cellulase and pectinase before electroporation.[2]

1. To remove cell wall to form protoplasts;
2. Protoplasts which are lacking in cell wall can then take up plasmid containing *Bt* gene during electroporation / cell wall is a barrier to uptake of *Bt* gene during electroporation;

(ii) Addition of mannitol to water lowers the water potential of water. In the preparation of protoplasts, mannitol must be present. In the absence of mannitol, no protoplasts are formed even if the sterilised explant was treated with cellulase and pectinase.

State what happens to the protoplasts if mannitol was not used. [1]

The protoplasts lysed;

(c) Name an alternative method of introducing the *Bt* gene into the plant cells. [1]

1. Gene gun / *Agrobacterium*-mediated gene transfer

(d) Explain the benefit of incorporating a *Bt* gene in the plasmid as shown in **Fig.4.1**. [2]

1. Increase crop yield;
2. Bt toxin is synthesized making crops resistant to insects that consume it/plant is pest resistant or kill the pest that consume it;

[Total : 8]

Section B

Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Explain the effects of competitive and non-competitive inhibitors on the rate of enzyme activity. [8]

Competitive inhibitor [max 4]

1. Competitive inhibitor of enzymes has a **structural resemblance to the substrate** thus competes with substrate for active site.
2. Competitive inhibitor is **complementary in shape to the active site** and thus it binds to the active site of the enzyme
3. It **binds reversibly** to the active site and the bonds involved are weak, non-covalent bonds
4. It **reduces the availability of enzyme active sites** for substrate binding and hence reduces the rate of the enzymatic reaction.
5. Inhibition can be **overcome by increasing substrate concentration**
6. which increases the chances of substrate binding to active site instead of inhibitor binding.
7. At sufficiently high [S], reaction velocity reaches the **same V_{max}** observed in the absence of inhibitor.

Non-competitive inhibitor [max4]

8. Non-competitive inhibitor **bears no structural similarity** to the substrate molecule so it binds to a site other than the enzyme active site

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9. It **alters the conformation of the enzyme active site** rendering it unreceptive to substrate molecules. This lowers the rate of reaction.
10. Non-competitive Inhibitor effectively **decreases the availability of enzymes** as it forms inactive enzyme-inhibitor complexes.
11. Effects of inhibitors **cannot be overcome** by high concentration of substrate
12. The **rate of reaction will continue to decrease** with increasing inhibitor concentration. When inhibitor saturation is reached, the rate of reaction will be almost zero

(b) Explain why variation is important in selection. [5]

1. Variation is the **differences between individuals of the same species due to the presence of different alleles**.
2. It includes morphological, physiological, biochemical and genetic differences.
3. It is the **raw material for natural selection to act on**.
4. If not for variation, s**election would not be possible as there would be no differences between individuals** of a population.
5. Variations arise spontaneously and are not dictated by the need of organisms to survive better in the environment.
6. Some variants are less adapted (due to the presence of unfavourable alleles) to survive in an environment compared to others. Other variants are better adapted to survive.
7. Variation **helps to ensure perpetuation of species** and safeguard species from extinction when the environment changes
8. When **environmental changes** such as a change in climate or an introduction of new predators, the variation allows some individuals **(i.e. those with favourable alleles, that confer an advantage)** to survive and reproduce more successfully than others.
9. Variations provide the basis for natural selection to select individuals with characteristics that are f**avoured by the environment**.

(c) Explain how natural selection may bring about evolution. [7]

1. There is **overproduction of offspring** as all organisms produce large number of offspring
2. Despite the tendency towards overpopulation, most populations show **constancy of numbers** / remain relatively **stable in size** because only a small fraction of the offspring population survives, matures and reproduces.
3. There is **struggle for survival** because of competition for finite resources such as food, soil nutrients, water, mates, habitats;
4. and other factors such as disease and predators impose a limit on their number
5. There is **variation within a population which is the raw material for natural selection to act on.**
6. There is **survival of the fittest** by natural selection as when environment changes, the variation allows some individuals with **favourable alleles** that **confer an advantage** to survive and reproduce more successfully than others.
7. These survivors have characteristics that are **selected**/**favoured by the environment**. The traits help them **adapt better** to the new environment. The organisms which survive get a chance to produce **viable and fertile offspring.**
8. **Like produces like as t**hose which survive breed to produce offspring **similar to themselves**
9. Advantageous characteristics (determined by **favourable alleles**) are more likely to be **passed on to the offspring**

10. With each succeeding generation, the proportion of individuals having the advantageous characteristics **(determined by favourable alleles)** increases while those lacking the characteristics decreases. Favourable characteristics, and hence favourable genotypes accumulate over time leading to f**ormation of a new species over a long period of time as**

11. Over hundreds and thousands of **generations** spanning about 200,000 years, a new species may form.

[Total: 20]

OR

6 (a) Compare DNA polymerase and *Taq* polymerase. [5]

Similarities (max 2)

1. Both cannot synthesize DNA from scratch, and require a free 3' –OH end to catalyze the synthesis of a new DNA strand.
2. Both require a template to catalyze the synthesis of a new DNA strand via complementary base-pairing.

Differences

Feature	DNA polymerase	Taq polymerase
3. Proofreading capability	DNA polymerase has 3'->5' proofreading capability.	No proofreading capability.
4. Optimum temperature / thermostability	37°C	72°C
5. Type of primer	Utilizes RNA primer during DNA replication in cells.	Utilizes DNA primer During PCR.

(b) Describe the properties of plasmids that allow them to be used as DNA cloning vectors. [6]

1. Has an origin of replication, multiple cloning site (MCS) and at least two selectable markers.
2. Origin of replication allows plasmid to replicate independently of bacterial chromosome.
3. This allows for multiple plasmids to exist within each bacterium.
4. In turn, this also increases the probability / ensures that progeny bacteria each receive a copy of the plasmid.
5. The MCS contains many different restriction sites.
6. This allows the plasmid to be used for cloning of a wide range of different genes / DNA fragment of interest.
7. e.g. of the 2 selectable markers: ampicillin resistance gene and tetracycline resistance gene / ampicillin gene and lacZ gene.
8. One of the selectable markers allow for the selection of bacteria transformed with the plasmid.
9. The other selectable marker allows for the selection of bacteria transformed with the recombinant plasmid / plasmid containing the inserted gene of interest.

(c) Outline how insulin can be produced by genetic engineering. [9]

1) Formation of ds cDNA:
2) Mature insulin mRNA is obtained from the beta cells of the islets of Langerhans.

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3) Using oligo-dT as a primer, mature insulin mRNA is reversed transcribed by reverse transcriptase to form a single stranded complementary DNA (cDNA).
4) RNase / NaOH treatment is used to remove the RNA from the DNA/RNA hybrid.
5) DNA polymerase then synthesizes the second cDNA strand, forming a double stranded cDNA molecule.

6) Formation of complementary sticky ends:
7) DNA linkers are added to the blunt ends of the ds cDNA and cleaved with a restriction enzyme to produce sticky ends.
8) The same restriction enzyme is used to cleave a plasmid to produce complementary sticky ends.
9) (note: alternatively terminal transferase method)

10) Formation of recombinant plasmid and transformation of bacterial cells:
11) The cDNA and plasmid are then mixed to allow for annealing to take place between complementary sticky ends through the formation of hydrogen bonds.
12) Ligase is used to seal the nicks through the formation of phosphodiester bonds between adjacent nucleotides, forming the recombinant plasmid.
13) Bacterial cells are made competent using $CaCl_2$ solution and transformed with the plasmid using heat shock.

14) Selection of recombinant bacterial cells:
15) Since plasmid contains 2 selectable markers (e.g. ampicillin resistance gene and lacZ gene), and lacZ gene is insertionally inactivated by the insulin cDNA,
16) white bacteria colonies with the ability to grow on ampicillin- and X-gal-containing media are the ones transformed with the recombinant plasmid and are selected.
17) (alternatively: plasmid containing amp^r and tet^r and description of replica plating)

18) Large-scale production and formation of functional insulin:
19) These recombinant bacteria are then grown in a fermenter and the proinsulin collected.
20) Proinsulin is then converted into functional insulin through the removal of the C chain and the formation of disulfide bonds between the A and B chains.

[Total: 20]